FakeImageDetector

Comprehensive Project Report

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1. Executive Summary

FakeImageDetector is a comprehensive deep learning system designed to detect fake, tampered, and AI-generated images. The project implements multiple detection methods including Error Level Analysis (ELA) combined with Convolutional Neural Networks (CNN), EfficientNet-B4 based AI-generated image detection, and metadata analysis. This system addresses the growing need for image authenticity verification in the age of advanced image manipulation tools and AI-generated content.

2. What is FakeImageDetector?

FakeImageDetector is a machine learning project that provides multiple approaches to detect fake and tampered images. The system consists of three main detection methods:

2.1 ELA + CNN Detection Method

This method combines Error Level Analysis (ELA) with Convolutional Neural Networks. ELA is a technique that reveals compression artifacts in images, which are different for original versus edited/tampered images. The ELA-processed images are then fed into a custom CNN architecture for classification as real or fake.

Key Features:

• Detects image tampering and splicing

• Uses ELA to highlight compression differences

• Custom CNN architecture for classification

• Achieves ~91.83% accuracy

2.2 AI-Generated Image Detection

This method uses transfer learning with EfficientNet-B4 to detect AI-generated images. The model can classify images into three categories: Real images, GAN-generated images (StyleGAN, ProGAN), and Diffusion model-generated images (DALL-E, Midjourney, Stable Diffusion).

Key Features:

• Uses pre-trained EfficientNet-B4 backbone

• Detects GAN-generated images

• Detects Diffusion model-generated images

• Custom classification head for fine-tuning

2.3 Metadata Analysis

This component analyzes EXIF metadata from images to detect inconsistencies that may indicate tampering or manipulation. It examines timestamps, creation dates, and other metadata attributes.

3. Why FakeImageDetector?

In today's digital age, image manipulation and AI-generated content have become increasingly sophisticated and prevalent. This creates several critical challenges:

3.1 Problem Statement

Fake News and Misinformation: Manipulated images are often used to spread false information, which can have serious consequences in journalism, social media, and public discourse.

Security Concerns: Fake images can be used for identity theft, fraud, and other malicious activities.

AI-Generated Content: With the rise of DALL-E, Midjourney, Stable Diffusion, and other AI image generators, it's becoming harder to distinguish between real and synthetic images.

Forensic Applications: Law enforcement and legal proceedings require reliable methods to verify image authenticity.

3.2 Solution Benefits

FakeImageDetector addresses these challenges by providing:

• **Multiple Detection Methods:** Uses different techniques to catch various types of manipulations

• **High Accuracy:** Achieves over 90% accuracy in detecting tampered images

• **AI Detection:** Specifically designed to detect modern AI-generated content

• **Comprehensive Analysis:** Combines visual analysis with metadata examination

• **Open Source:** Provides transparency and allows for community contributions

4. How Does It Work?

4.1 ELA + CNN Method

Step 1: Error Level Analysis (ELA)
The original image is saved at a specific JPEG quality (e.g., 90%). The difference between the original and re-saved image is computed. Edited regions show higher error levels, making tampering visible.

Step 2: Preprocessing
The ELA-processed image is resized to 128×128 pixels and normalized to values between 0 and 1.

Step 3: CNN Classification
The preprocessed image is fed through a CNN with multiple convolutional layers, pooling layers, and dense layers to classify it as real or fake.

4.2 AI-Generated Image Detection

Step 1: Image Preprocessing
Input images are resized to 224×224 pixels (EfficientNet input size) and normalized to [0, 1] range.

Step 2: Feature Extraction
The EfficientNet-B4 base model (pre-trained on ImageNet) extracts high-level features from the images.

Step 3: Classification
A custom classification head with dense layers and dropout classifies images into Real, GAN-generated, or Diffusion-generated categories.

4.3 Metadata Analysis

Extracts and analyzes EXIF metadata including timestamps, camera information, GPS data, and software used. Inconsistencies in metadata can indicate tampering.

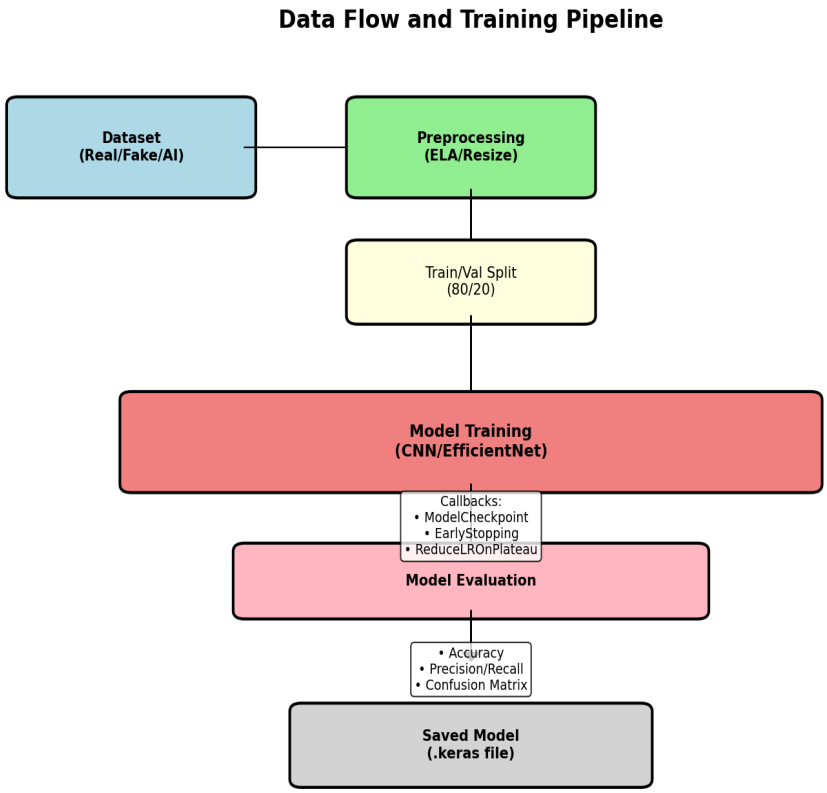

Figure 1: Data Flow and Training Pipeline

5. Architecture

5.1 System Overview

The system consists of three main detection pipelines that can work independently or in combination to provide comprehensive image authenticity verification.

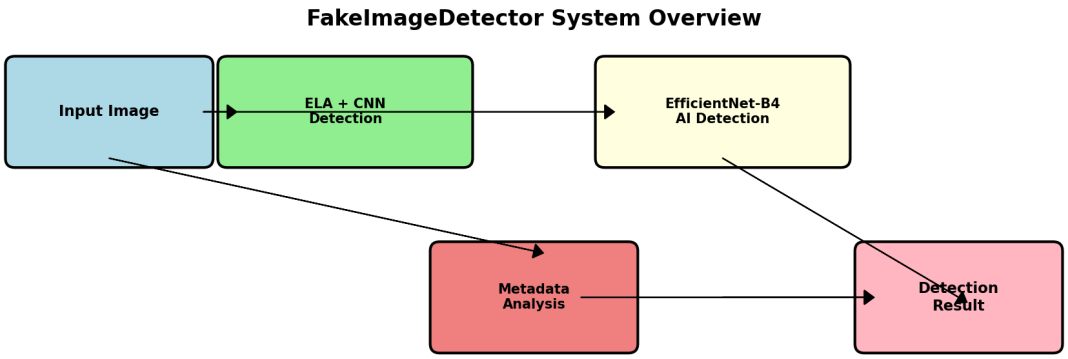

Figure 2: System Overview

5.2 ELA + CNN Architecture

The ELA+CNN architecture processes images through error level analysis and then uses a custom CNN for binary classification.

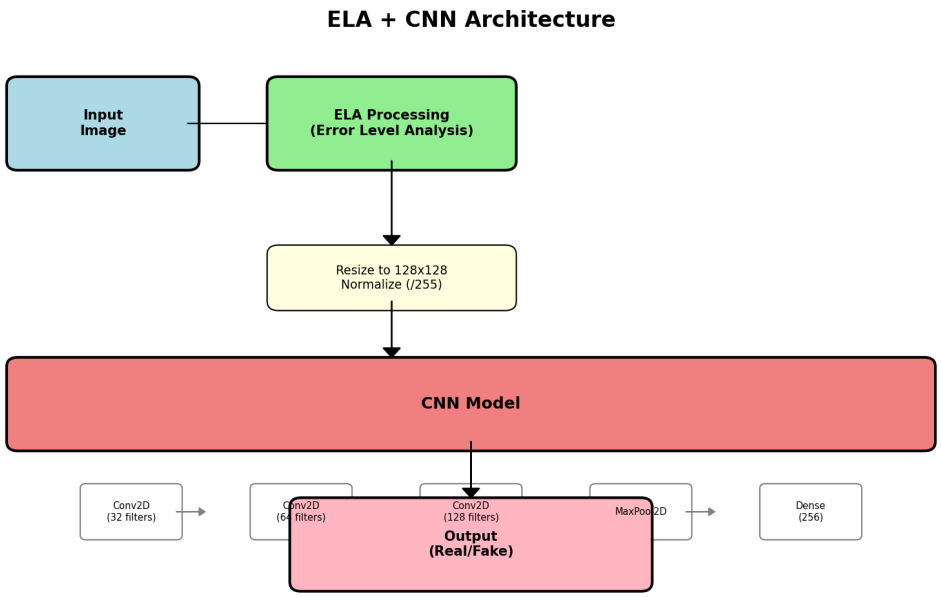

Figure 3: ELA + CNN Architecture

5.3 EfficientNet-B4 AI Detection Architecture

The AI detection model uses transfer learning with EfficientNet-B4, a state-of-the-art architecture that balances accuracy and efficiency. The model includes a custom classification head with dropout regularization.

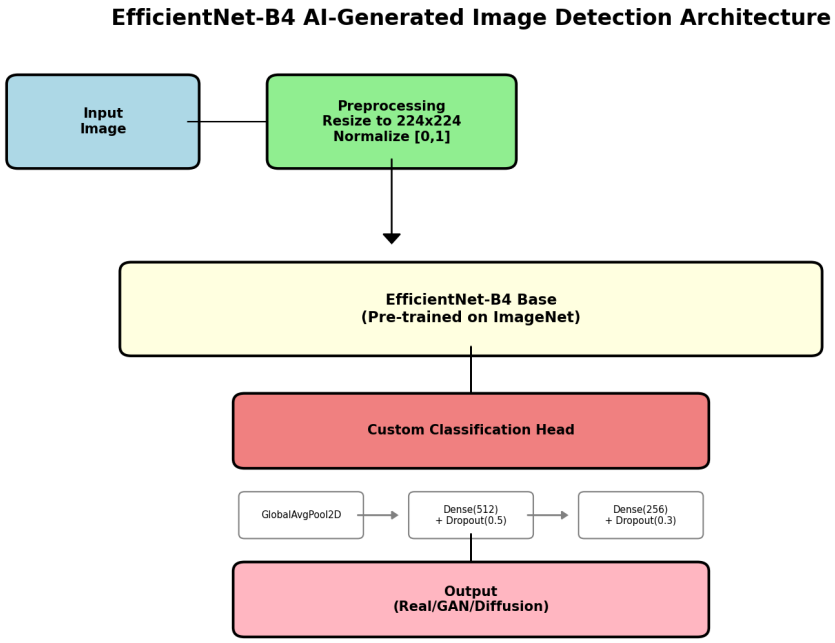

Figure 4: EfficientNet-B4 AI Detection Architecture

Model Details:

Component	Specification
Base Model	EfficientNet-B4 (pre-trained on ImageNet)
Input Size	224×224×3
Classification Head	GlobalAveragePooling2D → Dense(512) → Dropout(0.5) → Dense(256) → Dropout(0.3)
Output Classes	Real, GAN-generated, Diffusion-generated
Optimizer	Adam (learning rate: 0.0001)
Loss Function	Categorical Crossentropy

6. Dataset

6.1 ELA Detection Dataset

The ELA detection method uses a dataset of approximately 9,500 images split into real and fake categories. Images are processed through ELA before training.

6.2 AI Detection Dataset

The AI detection dataset consists of three categories:

Category	Count	Source
Real Images	~31,783 images	Natural photographs from sources like COCO dataset
GAN-Generated	~70,001 images	Images generated using StyleGAN, ProGAN, etc.
Diffusion-Generated	~2,778 images	Images from DALL-E, Midjourney, Stable Diffusion

```
Dataset Structure:
datasets/
■■■ ai_detection/
■ ■■■ real/
■ ■■■ gan/
■ ■■■ diffusion/
■ ■■■ test/
■■■ train/
■ ■■■ real/
■ ■■■ fake/
■■■ dataset.csv
```

7. Training Process

7.1 ELA + CNN Training

Configuration:
• Image size: 128×128×3 (after ELA processing)
• Batch size: 100
• Epochs: 30 (with early stopping)
• Optimizer: RMSprop (learning rate: 0.0005)
• Train/Validation split: 80/20

Training Process:
1. Load images from CSV file
2. Convert each image to ELA format
3. Resize and normalize
4. Split into training and validation sets
5. Train CNN model with early stopping callback
6. Save best model based on validation accuracy

Results: Model converged at epoch 9 with 91.83% accuracy.

7.2 AI Detection Training

Configuration:
• Image size: 224×224×3
• Batch size: 32
• Epochs: 50 (with early stopping)
• Optimizer: Adam (learning rate: 0.0001)
• Train/Validation split: 80/20 (stratified)

Training Process:
1. Load images from folder structure (real/gan/diffusion)
2. Preprocess: resize to 224×224, normalize to [0,1]
3. One-hot encode labels for multi-class classification
4. Load pre-trained EfficientNet-B4 weights
5. Fine-tune with custom classification head
6. Use callbacks: ModelCheckpoint, EarlyStopping, ReduceLROnPlateau
7. Save best model based on validation accuracy

Callbacks:
• **ModelCheckpoint:** Saves best model during training
• **EarlyStopping:** Stops training if no improvement for 5 epochs
• **ReduceLROnPlateau:** Reduces learning rate when validation loss plateaus

8. Results and Performance

8.1 ELA + CNN Method

The ELA + CNN model achieved excellent performance in detecting tampered images:

• **Best Accuracy:** 91.83%
• **Convergence:** Epoch 9
• **Method:** Binary classification (Real vs Fake)

8.2 AI Detection Method

The EfficientNet-B4 model provides multi-class classification for detecting different types of AI-generated content. Performance metrics include:

• **Classification:** 3-class (Real, GAN, Diffusion)
• **Model Size:** ~215 MB
• **Input Processing:** Supports various image formats (JPG, PNG, BMP)
• **Inference:** Can process single images or batches

8.3 Model Evaluation Metrics

The project includes comprehensive evaluation including:

• Classification reports with precision, recall, and F1-score
• Confusion matrices for visual analysis
• Training curves (loss and accuracy over epochs)
• Validation accuracy tracking

9. Project Structure

The project is organized as follows:

```
FakeImageDetector/
■■■ ai_generated_detection.ipynb      # EfficientNet-B4 AI detection notebook
■■■ fake-image-detection.ipynb        # ELA + CNN detection notebook
■■■ metadata_analysis.ipynb           # EXIF metadata analysis notebook
■■■ generate_diffusion_images.py      # Script to generate diffusion images
■■■ datasets/                         # Dataset directories
■   ■■■ ai_detection/
■   ■   ■■■ real/                      # Real images
■   ■   ■■■ gan/                       # GAN-generated images
■   ■   ■■■ diffusion/                 # Diffusion-generated images
■   ■   ■■■ test/                      # Test images
■   ■■■ train/
■   ■   ■■■ real/                      # Real training images
■   ■   ■■■ fake/                      # Fake training images
■   ■■■ dataset.csv                    # Dataset metadata
■■■ models/                            # Trained models
■   ■■■ ai_generated/
■   ■   ■■■ ai_detector_best.keras
■   ■   ■■■ ai_detector_final.keras
■   ■   ■■■ training_history.pkl
■   ■■■ fake_image_detector_model.keras
■   ■■■ training_history.pkl
■■■ docs/                              # Documentation
■   ■■■ model-architecture.jpg
■   ■■■ Deteksi Pemalsuan Gambar dengan ELA dan Deep Learning.pdf
■■■ AI_DETECTOR_TRAINING_STEPS.md      # Training guide for AI detection
■■■ AI_DETECTION_DATASET_GUIDE.md      # Dataset preparation guide
■■■ TRAINING_GUIDE.md                  # Training guide for ELA detection
■■■ README.md                          # Project readme
■■■ LICENSE                            # License file
```

9.1 Key Files

Notebooks:

• **ai_generated_detection.ipynb:** Main notebook for AI-generated image detection using EfficientNet-B4

• **fake-image-detection.ipynb:** Notebook for ELA-based tampering detection

• **metadata_analysis.ipynb:** Notebook for EXIF metadata analysis

Documentation:

• Comprehensive training guides for both detection methods

• Dataset preparation and organization guides

• Architecture documentation

10. Conclusion

FakeImageDetector provides a comprehensive solution for detecting fake, tampered, and AI-generated images. The project successfully combines multiple detection techniques:

Key Achievements:
• Implemented ELA + CNN method achieving 91.83% accuracy
• Developed EfficientNet-B4 based AI detection for modern synthetic images
• Created comprehensive metadata analysis tools
• Provided detailed documentation and training guides

Applications:
• Journalism and fact-checking
• Social media content verification
• Digital forensics
• Security and authentication systems
• Research and education

Future Enhancements:
• Integration of multiple detection methods for ensemble predictions
• Real-time detection API
• Mobile application deployment
• Expanded dataset for better generalization
• Additional detection methods (PRNU, frequency domain analysis)

This project demonstrates the effectiveness of combining traditional image forensics techniques with modern deep learning approaches to address the growing challenge of image authenticity verification in the digital age.